

SIT225 Data Capture Technologies

High Distinction Task

System Hardening, Validation & Engineering Justification

Estimated time: 10-15 hours

Last updated: 14 June 2026

Due: End of Week 11, no later than week 12

Submit early, by week 10-11, to receive feedback early.

I. Instructions and Conditions

This task is due in **Week 11 (Sunday)**, and you are expected to start early. If your first submission is incomplete or incorrect, you may resubmit after receiving feedback. There will be **no extensions**, and late submissions will not be accepted. This task is a High-Distinction task, failure to submit a correct version on time will result in not availing a HD grade in the unit.

All submissions will be checked for **plagiarism** and **GenAI** use. You must complete the work **independently**, and you must not share or show your work to anyone else, unless it is explicitly instructed in your task to do so. If you have questions or encounter problems, attend the on-campus or online classes, or use the **Discussion Board** on the unit site.

All submissions will be checked for **plagiarism** and **GenAI** use. *Any GenAI usage should be acknowledged (see template below) and you must include, in appendix, the prompt, raw response, and your changelog how you have adopted the GenAI response in your work.* You must complete the work **independently**, and you must not share or show your work to anyone else, unless it is explicitly instructed in your task to do so. If you have questions or encounter problems, attend the on-campus or online classes, or use the **Discussion Board** on the unit site.

Per Deakin policy, include an Acknowledgements section:

"This assessment work was developed with assistance from the following approved Deakin genAI tools:

- Deakin GEM (accessed June 2026): Used to *<generate structured critique on protocol design and confounding variables (3 iterations)>*.
- Microsoft Copilot for web (accessed June 2026): Used to *<summarise ASHRAE comfort standards and identify relevant literature.>*

All genAI-generated suggestions were critically evaluated and integrated only where supported by evidence. The final design, decisions, and justifications are my own."

II.Task Overview

You will transform your prototype into a robust, production-quality system candidate, supported by rigorous validation, testing, and engineering justification. You must demonstrate that your solution is not only functional but also dependable, trustworthy, and professionally defensible.

III.Task Components

Part 1: Advanced Testing & Quality Assurance

Design and execute a comprehensive testing strategy, including:

- Unit testing (key modules)
- Integration testing (sensor -> processing -> output)
- Edge-case testing (invalid/missing sensor data)
- Performance testing (latency, throughput)
- Reliability testing (stability over time)

HD Level:

- Uses systematic test plan with clear coverage
- Includes automated testing where possible
- Reports quantitative metrics (e.g., response time, error rates)

Part 2: System Validation Against Real-World Objectives

Revisit the original problem (Task 1 or 3, which one is appropriate) and validate:

- Does the system accurately detect meaningful events?
- Are alerts useful and actionable?
- Does it reduce or mitigate the identified problem?

HD Level:

Evidence-based validation:

- Data comparisons (baseline vs system outcomes)
- Measurable success criteria

Identifies:

- False positives / negatives
- Limitations of sensing or thresholds

Part 3: System Robustness & Fault Tolerance Design

Extend the system to handle real-world uncertainties:

- Sensor failure handling
- Network interruptions
- Noisy or inconsistent data

HD Level:

- Implements or simulates:

- Error handling strategies
- Data smoothing / filtering
- Demonstrates graceful degradation

Part 4: Software Architecture & Engineering Justification

Produce a formal architecture description, including:

- Component diagram
- Data flow
- Technology stack justification

HD Level:

- Explains:
 - Why this architecture fits the problem
 - Trade-offs (e.g., simplicity vs scalability)
- Links decisions to:
 - Performance
 - Maintainability
 - Ethical constraints (e.g., local vs cloud processing)

Part 5: Operational Ethics & Risk Mitigation

Move beyond theoretical ethics (Task 2) into practical safeguards:

- Data minimisation strategies
- Privacy-preserving design (e.g., anonymisation)
- Risk register (technical + social risks)

HD Level:

- Provides:
 - Concrete mitigation mechanisms
 - Realistic risk scenarios
- Demonstrates professional responsibility mindset

Part 6: Professional Engineering Report

Produce a structured, evidence-driven report synthesising all tasks, as detailed in the submission checklist section (section IV) below.

Suggested Sections:

1. Problem & Context Recap
2. System Evolution (Tasks 1-4 summary)
3. Testing & Validation Results
4. Architecture & Engineering Decisions
5. Ethical & Risk Considerations

6. Limitations & Future Improvements

HD Level:

- Clear traceability across all tasks
- Uses data, figures, and evidence
- Written to industry/professional standard

Part 7: Conference article draft

At this point, you should be confident on the work you have done so far and would prefer to share with the world through a scholarly conference paper (*max 4 pages in 2 column format, see a template link in submission checklist below*). You can seek assistance from the tutors in preparing this draft and if time permits, continue updating the draft by including additional results, analysis and performance comparison to submit in a future conference. The draft can be perceived as a coherent and concise story around the problem, method and solution; you should be reusing part 6 report content as required.

Suggested sections:

1. Introduction (3-4 short paragraphs including problem statement, background, and your contribution)
2. Related works (baseline references you explored in assessment 2)
3. Methods
4. Results & Discussion
5. Conclusion

HD Level:

- Clear and concise draft with an engaging story line
- Image diagram showing your method description
- Results presented in tables
- A performance comparison table in the Discussion
- Any limitation and future work

What Makes This High-Distinction

A high-distinction submission will:

- Demonstrate engineering depth beyond prototype development
- Provide quantifiable evidence of system quality and effectiveness
- Show robustness in handling real-world uncertainty
- Justify design decisions with technical and ethical reasoning
- Communicate like a professional software engineer
- Produce a scholarly output to share with the research community

Example High-Distinction Insight:

While initial testing showed 85% alert accuracy, stress testing under fluctuating sensor noise revealed instability. Introducing a Kalman filter reduced false positives by 27%, improving user trust without increasing latency.

One-Line Summary:

Elevate your prototype into a professionally engineered system by rigorously testing, validating, and justifying its robustness, effectiveness, and real-world readiness.

IV.Submission Checklist & HD Grading Criteria

Component	Format	Criteria
Part 1: Testing Strategy & Quality Assurance (20%)	--	Comprehensive, well-structured testing strategy (unit, integration, edge cases, performance). Automated testing or reproducible tests included. Clear coverage justification and quantitative metrics (e.g., latency, error rates).
Part 2: System Validation & Effectiveness (20%)	--	Rigorous validation against original problem. Uses baseline comparisons, measurable success criteria, and clear evidence. Critically analyses false positives/negatives and system limitations.
Part 3: Robustness & Fault Tolerance (15%)	--	Demonstrates advanced handling of real-world uncertainty (sensor failure, noisy data, missing values). Implements mitigation strategies (e.g., filtering, fallback logic) with clear justification.
Part 4: Software Architecture & Technical Justification (15%)	--	Clear, professional architecture (diagrams + explanation). Strong justification of design decisions and trade-offs (e.g., scalability, maintainability, privacy).
Part 5: Ethics, Risk & Professional Responsibility (15%)	--	Advanced ethical analysis applied in practice. Includes data governance, privacy safeguards, and a well-developed risk register with mitigation strategies. Demonstrates professional responsibility.
Part 6: Professional Engineering Report (15%)	--	Exceptionally clear, well-structured report. Strong integration across Tasks 1-4. Uses evidence (figures, data) effectively. Professional tone and presentation.
Part 7: Conference article draft	--	A PDF 4 page 2 column report ready for a conference submission. IEEE conference template (https://www.ieee.org/conferences/publishing/templates)
Optional Bonus (up to +5%)		For exceptional work: • Advanced techniques (e.g., ML-based detection, adaptive thresholds) • Comparative evaluation with existing systems • Deployment-ready considerations (cloud/edge trade-offs)

Consolidate each part of the task including any code/data (you can upload CSV data, if any, in OnTrack chat) into a single PDF.